

BORING LOG

Project: Sage Creek Canal — Reach 7 Rehabilitation
Location: Canal invert, Sta. 153+00 (slide area)
Ground Elev: 4,285.0 ft
Drilling: CME-75, 4.25-in HSA

BH-SC-01

HDE Project No. 2025-IR-0012
Date: January 8-9, 2025
Total Depth: 30 ft
Driller: Snake River Drilling, LLC

Groundwater: 8.5 ft bgs

Depth (ft)	Graphic	USCS N-value	Material Description	Samples / Remarks	
0		CONCRETE LINING — Remnant, cracked and displaced by slide. 2.5-3.5 in thick.			Removed by auger
3					
		ML	N=10 N=14 N=16	LOESS — Tan, wind-deposited silt, loose to medium stiff. Slightly moist to moist. Collapse potential 8.5%. Open structure, visible macropores.	SS-1 to SS-3 CP-1 at 3-5 ft FH-1 at 4-6 ft
10					
		SP	N=8	ALLUVIAL SAND — Yellow-brown, poorly graded fine to medium sand, loose, saturated. Seepage conduit.	SS-4 Slug test 10-14 ft
12					
		CH	N=18 N=22 N=25 N=28	LACUSTRINE CLAY — Green-gray to olive, fat clay, very stiff to hard. LL 58-62, PI 34-37. Slickensided surfaces. Calcareous nodules. Swell pressure 4,200-5,800 psf. Highly expansive.	SS-5 to SS-8 ST-1 at 13-15 ft ST-2 at 18-20 ft DS-2 at 14-16 ft Slug test 16-20 ft
24					
		Refusal at 24 ft	BASALT BEDROCK — Dark gray, vesicular to massive olivine basalt. Upper 2 ft vesicular and fractured. Auger refusal.		No samples (auger refusal)
30					
TD					

BORING LOG

BH-SC-02

Project: Sage Creek Canal — Reach 7 Rehabilitation

Location: Right bank crest, Sta. 153+00 (slide head scarp)

Ground Elev: 4,297.0 ft

Drilling: CME-75, 4.25-in HSA

Groundwater: 14.0 ft bgs

HDE Project No. 2025-IR-0012

Date: January 10-12, 2025

Total Depth: 40 ft

Driller: Snake River Drilling, LLC

Depth (ft)	Graphic	USCS N-value	Material Description	Samples / Remarks
0		TOPSOIL — Brown silty loam, roots, organic matter. Agricultural surface.		
2				
		ML/CL N=8 N=12 N=10 N=15	LOESS — Tan to brown, wind-deposited silt to silty clay. Medium stiff when dry, collapses when wet. Collapse potential 6.8-10.2%. Tension cracks (open, 0.5-1 in) from 2-6 ft depth. Wet below 8 ft.	SS-1 to SS-4 CP-1 at 4-6 ft CP-2 at 8-10 ft DS-1 at 6-8 ft FH-1 at 6-8 ft
14 ^W		CH N=15 N=20 N=24 N=26 N=28	LACUSTRINE CLAY — Green-gray, fat clay, stiff to very stiff. LL 55-68, PI 32-42. Prominent slickensided surfaces at 16-18 ft (failure plane of right bank slide). Calcareous nodules. Swell pressure 3,600-6,500 psf.	SS-5 to SS-10 ST-1 at 15-17 ft ST-2 at 22-24 ft DS-2 at 16-18 ft DS-3 at 24-26 ft
32		SP-SM N=12	ALLUVIAL SAND — Brown, poorly graded sand with silt, loose to medium dense, saturated.	SS-11
35		Refusal at 35 ft	BASALT BEDROCK — Dark gray, vesicular basalt grading to massive. Auger refusal at 35 ft. Boring terminated.	
40				
TD				

BORING LOG

Project: Sage Creek Canal — Reach 7 Rehabilitation
Location: Canal invert, Sta. 149+00 (lining heave area)
Ground Elev: 4,286.5 ft
Drilling: CME-75, 4.25-in HSA

Groundwater: 9.0 ft bgs

BH-SC-03

HDE Project No. 2025-IR-0012
Date: January 13, 2025
Total Depth: 25 ft
Driller: Snake River Drilling, LLC

Depth (ft)	Graphic	USCS N-value	Material Description	Samples / Remarks
0			CONCRETE LINING — Heaved panels, 3-4 in thick, cracked at midspan. Upward displacement 3-5 in. ASR gel visible.	CC-3 core extracted from adjacent panel
2				
		ML N=12 N=15	LOESS — Tan silt, medium stiff, slightly moist to moist. Collapse potential 7.5%. Open macropore structure.	SS-1 to SS-3 CP-1 at 3-5 ft FH-1 at 4-6 ft PD-1
		SP N=6	ALLUVIAL SAND — Fine to medium sand, loose, saturated. Seepage observed during drilling.	SS-4 Slug test 8-12 ft
11				
		CH N=16 N=20 N=24	LACUSTRINE CLAY — Olive to green-gray, fat clay, very stiff. LL 60-64, PI 35-40. Slickensided. Swell pressure 4,800-5,500 psf. This unit directly responsible for lining heave.	SS-5 to SS-7 ST-1 at 11-13 ft ST-2 at 16-18 ft DS at 12-14 ft FH-2 PD-2
22				
		Refusal at 22 ft	BASALT BEDROCK — Vesicular to massive basalt. Auger refusal.	
25 TD				

BORING LOG

Project: Sage Creek Canal — Reach 7 Rehabilitation
Location: Left bank crest, Sta. 149+00 (stable control)
Ground Elev: 4,292.5 ft
Drilling: CME-75, 4.25-in HSA

Groundwater: 12.5 ft bgs

BH-SC-04

HDE Project No. 2025-IR-0012
Date: January 14, 2025
Total Depth: 20 ft
Driller: Snake River Drilling, LLC

Depth (ft)	Graphic	USCS N-value	Material Description	Samples / Remarks
0	TOPSOIL — Brown silty loam, agricultural surface.			
2				
		ML N=12 N=16 N=18	LOESS — Tan silt, medium stiff to stiff. Dry, no tension cracks. Collapse potential 5.5% (moderate). Intact structure.	SS-1 to SS-3 CP-1 at 3-5 ft DS-1 at 4-6 ft
10				
		CH N=18 N=22 N=25	LACUSTRINE CLAY — Green-gray, fat clay, very stiff. LL 57, PI 34. Slickensided surfaces less prominent than BH-SC-02. Swell pressure 3,800 psf.	SS-4 to SS-6 ST-1 at 10-12 ft DS-2 at 10-12 ft
18				
		Refusal at 18 ft	BASALT BEDROCK — Dark gray basalt, massive. Auger refusal.	
20				
TD				